OPERATING SYSTEMS / MOCK EXAM 06

180 minutes | 100 points | Name: _____

Show your work. Use the exact model and event rules specified in each question.

1. FIFO 1 - arrivals and an idle gap (5 points)

A: arrival=2, bursts=[5]; B: arrival=3, bursts=[2]; C: arrival=4, bursts=[4]; D: arrival=15, bursts=[1].

All bursts are CPU only. Ready events append in time order; simultaneous events use alphabetical order. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

2. FIFO 2 - overlapping I/O (5 points)

A: arrival=0, bursts=[2, 5, 2]; B: arrival=1, bursts=[4]; C: arrival=3, bursts=[2, 2, 1].

Burst lists alternate CPU, I/O, CPU; I/O runs independently and may overlap. Ready events append in time order; simultaneous events use alphabetical order. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

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3. SJF 1 - arrivals during a long job (5 points)

A: arrival=0, bursts=[6]; B: arrival=1, bursts=[5]; C: arrival=2, bursts=[2]; D: arrival=7, bursts=[1].

All bursts are CPU only. Ties use earlier original arrival, then alphabetical name. Nonpreemptive; choose the shortest available next CPU burst immediately. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

4. SJF 2 - shortest next burst after I/O (5 points)

A: arrival=0, bursts=[3, 4, 2]; B: arrival=0, bursts=[5]; C: arrival=1, bursts=[2, 3, 1].

Burst lists alternate CPU, I/O, CPU; I/O runs independently and may overlap. Ties use earlier original arrival, then alphabetical name. Nonpreemptive; choose the shortest available next CPU burst immediately. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

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5. STCF 1 - chained preemptions (5 points)

A: arrival=0, bursts=[10]; B: arrival=2, bursts=[5]; C: arrival=3, bursts=[2]; D: arrival=6, bursts=[1].

All bursts are CPU only. Ties use earlier original arrival, then alphabetical name. Preempt only for a strictly shorter remaining current CPU burst. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

6. STCF 2 - I/O return competes with arrivals (5 points)

A: arrival=0, bursts=[3, 4, 2]; B: arrival=1, bursts=[8]; C: arrival=4, bursts=[2].

Burst lists alternate CPU, I/O, CPU; I/O runs independently and may overlap. Ties use earlier original arrival, then alphabetical name. Preempt only for a strictly shorter remaining current CPU burst. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

7. RR 1 - I/O and boundary arrivals (5 points)

A: arrival=0, bursts=[3, 3, 2]; B: arrival=1, bursts=[5]; C: arrival=2, bursts=[2].

Burst lists alternate CPU, I/O, CPU; I/O runs independently and may overlap. Ready events append in time order; simultaneous events use alphabetical order. Quantum=2. At a boundary, arrivals/I/O completions enter before the expired job is requeued. Dispatch cost is zero.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

8. RR 2 - paid dispatch intervals (5 points)

A: arrival=0, bursts=[5]; B: arrival=1, bursts=[3]; C: arrival=4, bursts=[2].

All bursts are CPU only. Ready events append in time order; simultaneous events use alphabetical order. Quantum=2. At a boundary, arrivals/I/O completions enter before the expired job is requeued. A direct change of job costs 1 time unit; no cost for initial dispatch, after idle, or continuing the same job. The selected job is fixed during this cost; new events join the queue.

(a) Give the full CPU timeline, including idle/dispatch intervals. (3 points)

(b) Give turnaround and response for every job and both averages. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

9. MLFQ 1 - higher-queue arrival (5 points)

One CPU. Q0>Q1>Q2, with quantums 2,4,8. A new job enters Q0. Exhausting a quantum demotes one level; Q2 stays Q2. Same-level arrivals never preempt. A higher-level arrival preempts immediately; the interrupted job resumes at the front of its queue with its unused quantum. No I/O, boosts, or costs. A(0,9), B(1,3), C(5,2) are (arrival, CPU time).

(a) Give the full timeline and every demotion time. (3 points)

(b) Give all completion times and mean turnaround and response. (2 points)

10. MLFQ 2 - a specified priority boost (5 points)

One CPU; Q0>Q1. Q0 quantum=2; exhaustion demotes to Q1. Q1 quantum=4; exhaustion appends to Q1. All new jobs enter Q0; same-level jobs use RR. A(0,10) and B(0,6) arrive in A,B order. At time 8 only, first account for execution ending at 8, then move every unfinished job to Q0 in alphabetical order and reset its quantum. No I/O or overhead.

(a) Give the timeline, marking the boost at 8. (3 points)

(b) Give completion times, turnaround times, and mean response. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

11. Lottery 1 - a blocked job and new tickets (5 points)

A,B,C initially hold 20,30,50 tickets. Each lottery independently selects one runnable ticket uniformly, and grants 2 ms of useful CPU time. Phase I: all run for 40 lotteries. Phase II: A is blocked for 20 lotteries; only B,C participate with unchanged tickets. Phase III: A returns with 40 tickets, B,C keep 30,50, and all run for 60 lotteries. Ignore overhead.

(a) Give each job's expected wins in every phase. (3 points)

(b) Give each job's expected total CPU time and its fraction of the 240 ms total. (2 points)

12. Lottery 2 - transfer and currency conversion (5 points)

Group G has 60 global tickets, split between G1:G2 in local ratio 1:2. H has 40 global tickets. In Phase I, all run for 50 independent lotteries. In Phase II, G1 blocks and transfers all its effective global tickets to H; G2 keeps its allocation. G1 never participates in the next 30 lotteries. Each win grants 3 ms; overhead is zero.

(a) Convert local allocations to global tickets; give Phase I and II expected wins per job. (3 points)

(b) Give total expected CPU time for G1,G2,H and H's expected share of the full run. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

13. Stride 1 - unequal initial passes (5 points)

Stride=600/tickets. Every selection gives 1 ms, then adds stride to the selected pass. Always choose the smallest pass; ties use alphabetical order. A,B,C have tickets (50,30,20) and initial passes (24,0,30). No other changes occur.

(a) Give selections 1 through 10, and the pass vector after each selection. (3 points)

(b) Give final passes and total CPU time received by each job. (2 points)

14. Stride 2 - a new runnable job (5 points)

Stride=600/tickets. Every selection gives 1 ms, then adds stride to the selected pass. Always choose the smallest pass; ties use alphabetical order. A,B start with tickets (40,20) and passes (30,30). Immediately before selection 5, C arrives with 30 tickets and pass equal to the minimum current pass of A,B. No other changes occur.

(a) Give selections 1 through 9, and the pass vector after each selection. (3 points)

(b) Give final passes and total CPU time received by each job. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

15. CFS 1 - unequal virtual starting points (5 points)

Use a simplified CFS model: choose minimum vruntime, ties alphabetical. Every selected job runs exactly 3 ms, then vruntime increases by (1024/weight)*3. No preemption inside a slice, no I/O or overhead. Weights (A,B,C)=(1024,2048,512); initial vruntimes=(6,9,3).

(a) Give selections 1 through 9 and the vruntime vector after each selection. (3 points)

(b) Give final vruntimes and total real CPU time per job. (2 points)

16. CFS 2 - a weight change at a boundary (5 points)

Use a simplified CFS model: choose minimum vruntime, ties alphabetical. Every selected job runs exactly 3 ms, then vruntime increases by (1024/weight)*3. No preemption inside a slice, no I/O or overhead. Initial weights (A,B,C)=(1024,1024,2048), vruntimes=(0,0,0). Immediately before selection 5, B changes weight to 2048. Keep its accumulated vruntime unchanged.

(a) Give selections 1 through 10 and the vruntime vector after each selection. (3 points)

(b) Give final vruntimes and total real CPU time per job. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

17. Amdahl 1 - serial fraction and overhead (5 points)

A fixed job takes 200 s on one core. Ignoring all overhead, its measured speedup on 4 cores is 2.5. Infer the serial fraction from this measurement. For an actual run on N>1 cores, add elapsed overhead 0.5(N-1) s to the ideal Amdahl time.

(a) Find the serial fraction and actual elapsed times for N=8 and N=16. (3 points)

(b) Find both actual speedups and select the faster of those two runs. (2 points)

18. Amdahl 2 - two scaling regions (5 points)

On one core, a fixed job takes 300 s: 10% is serial, 60% divides over all N cores, and 30% divides over at most 4 cores. Ignore overhead. An optimization halves only the serial elapsed time; all other work is unchanged. Always use the original 300 s as the speedup baseline.

(a) Find elapsed time and speedup on N=8 before and after optimization. (3 points)

(b) Find the infinite-core speedup bounds before and after optimization. (2 points)

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Show your work. Use the exact model and event rules specified in each question.

19. Multiprocessor 1 - stealing with a cost (5 points)

Two private FIFO queues: CPU0=[A(9),B(5),C(3)], CPU1=[D(4),E(2)]. All jobs arrive at 0; jobs are nonpreemptive. No other overhead. Policy X never migrates. Policy Y: when CPU1 first becomes empty, it steals the last not-yet-started job C from CPU0. The steal costs CPU1 2 time units, then C runs. No further steals occur. CPU0 never pauses for the steal.

(a) Give both CPU timelines under X and Y. (3 points)

(b) Give each policy's makespan and useful utilization, defined as total useful CPU time/(2*makespan). (2 points)

20. Multiprocessor 2 - SQMS versus fixed queues (5 points)

Three cores, all arrivals at 0, nonpreemptive jobs. Policy S uses one FIFO queue [A(9),B(5),C(4),D(6),E(3),F(2)]; each idle core immediately takes the head job, with simultaneous dispatches in CPU0,CPU1,CPU2 order. Queue access cost is zero. Policy M uses fixed queues CPU0=[A,D], CPU1=[B,E], CPU2=[C,F], with no migration or overhead.

(a) Give all three CPU timelines under S and M. (3 points)

(b) Give each makespan and aggregate idle time over [0,makespan). (2 points)